

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: I/O operation on closed file
```

BytesIO is pretty similar to StringIO class in a way that it provides us with an in-memory stream for file operations but using a bytes buffer. The usage is quite similar to the StringIO module, except a few extra methods and operations.

BytesIO can be initialized using a vanilla constructor or by providing a bytes-like object to the same. Anything apart from the aforementioned two methods will raise a TypeError.

```
>>> b=io.BytesIO(b"abcdef") #Prefix 'b' represents a bytes-
like object
>>> b.getvalue()
b'abcdef'
>>> b=io.BytesIO("abcdef")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: a bytes-like object is required, not 'str'
```

It provides us with a `getbuffer()` method, which returns a read-write memoryview of the buffer. It should be noted that any changes done on the memoryview will correspondingly update the buffer too. Any attempt to close or resize the BytesIO object while the view is still active, will result in a BufferError. In order to close or resize the stream we need to release the view, only then can we perform any modifications.

```
>>> view = b.getbuffer()
>>> b.close()
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
BufferError: Existing exports of data: object cannot be
re-sized
>>> view.release()
>>> b.close() #the stream is successfully closed
```

timeit

`timeit` is an inbuilt python module to test execution times of relatively smaller code snippets. The `timeit` module evaluates the time by executing the code block multiple times so that we are provided with the most relevant